RFC-9901: LOCALSHIFT CORE CONVERSION ARCHITECTURE

1. Introduction

This specification defines the offline file conversion architecture for LocalShift.
The system is engineered for deterministic, privacy-first local processing on Windows.

1.1 Design Goals

The conversion pipeline adheres to three primary design goals:

- 100% offline local CPU execution without network dependencies.
- Preservation of original document layout and character semantics.
- Cooperative cancellation and strict memory bounds.

2. Adapter Lifecycle and Routing

All document conversions are mediated by the ConversionAdapter protocol.

2.1 Preflight Validation

Before execution, the planner verifies the existence, format, and size of the input.

2.2 Resource Attribution

Process execution is monitored for memory footprint and page count limits.
Maximum pages processed per document shall not exceed 2000 pages.

3. Security Considerations
The document converter enforces atomic temporary file writes to prevent data loss. Input files larger than 250 MiB are rejected prior to rasterization.

4. References
- ISO 32000-1: Document Management — Portable Document Format.
- Open Packaging Conventions (ECMA-376 Part 2).